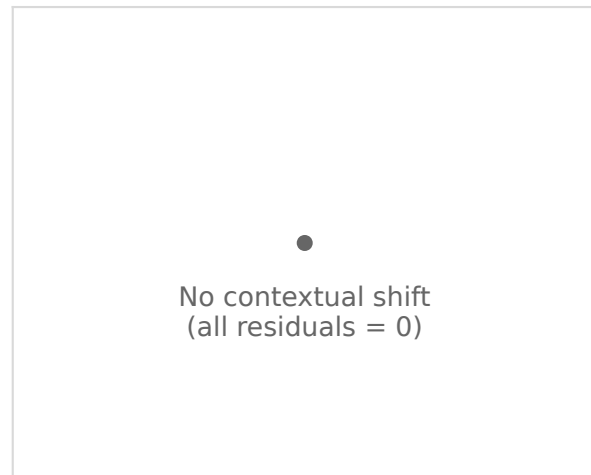


Gene-level contextual shifts · UMAP

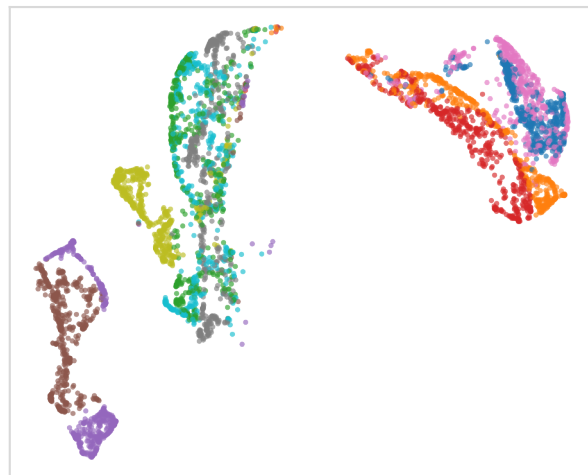
500 fixed genes per context · two representative samples per context

GTEx

L0 · static



L1 · early



L6 · middle



L12 · final

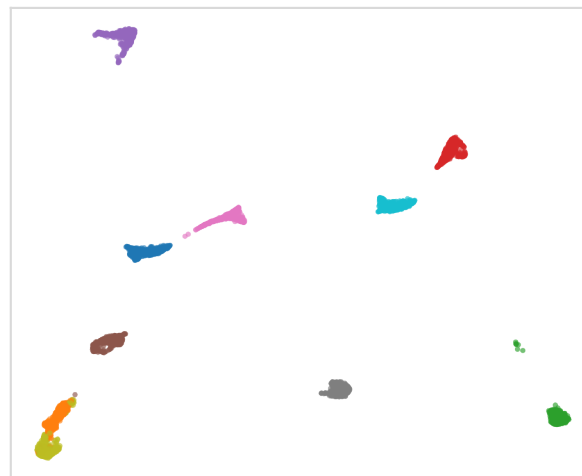
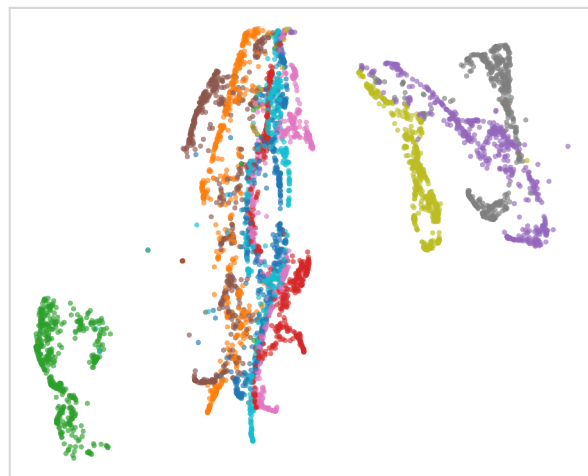
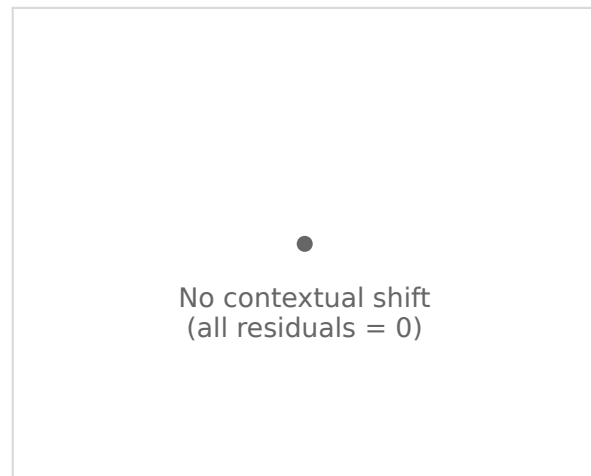


GTEx context

- heart
- skeletal muscle
- lung
- liver
- cortex
- cerebellum
- pancreas
- spleen
- testis
- subcutaneous adipose

TCGA

No contextual shift
(all residuals = 0)



TCGA context

- BRCA
- COAD
- GBM
- KIRC
- LIHC
- LUAD
- PRAD
- SKCM
- STAD
- THCA